Integration

These are ordered in (roughly) ascending difficulty.

Symmetry

Problem 1

Evaluate

$$\int_0^{\pi} \frac{\sin^{30}(x)}{\sin^{30}(x)+\cos^{30}(x)}dx.$$

Solution 1

Use Queens rule such that

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\sin^{30}(x)}{\sin^{30}(x)+\cos^{30}(x)}dx + \int_0^{\pi/2} \frac{\sin^{30}(\pi-x)}{\sin^{30}(\pi-x)+\cos^{30}(\pi-x)}dx \\ &= 2\int_0^{\pi/2} \frac{\sin^{30}(x)}{\sin^{30}(x)+\cos^{30}(x)}dx. \end{aligned}$$

Now we can use kings rule such that

$$I = 2\int_0^{\pi/2} \frac{\cos^{30}(x)}{\sin^{30}(x)+\cos^{30}(x)}dx$$

and so

$$\begin{aligned} 2I &= 2\int_0^{\pi/2} \frac{\sin^{30}(x)\cos^{30}(x)}{\sin^{30}(x)+\cos^{30}(x)}dx \\ &= \pi \end{aligned}$$

hence $I = \pi/2$.

Problem 2

Evaluate

$$\int_0^{2\pi} \frac{1}{1+e^{\sin(x)}}dx.$$

Solution 2

We use kings rule such that

$$\begin{aligned} I &= \int_0^{2\pi} \frac{1}{1+e^{\sin(2\pi-x)}}dx \\ &= \int_0^{2\pi} \frac{1}{1+e^{-\sin(x)}}dx \\ &= \int_0^{2\pi} \frac{e^{\sin(x)}}{1+e^{\sin(x)}}dx \end{aligned}$$

and so

$$\begin{aligned} 2I &= \int_0^{2\pi} \frac{1+e^{\sin(x)}}{1+e^{\sin(x)}}dx \\ &= 2\pi \end{aligned}$$

hence $I = \pi$.

Floor and Ceiling Functions

Problem 3

Evaluate

$$\int_0^{\infty} \lfloor x \rfloor e^{-\lfloor x \rfloor} dx.$$

Solution 3

$$\begin{aligned} I &= \sum_{n=0}^{\infty} \int_n^{n+1} ne^{-n} dx \\ &= \sum_{n=0}^{\infty} ne^{-n} \end{aligned}$$

Now we note that $0 < e^{-1} < 1$ and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

meaning that

$$\begin{aligned} \frac{1}{(1-x)^2} &= \frac{d}{dx}\frac{1}{1-x} \\ &= \frac{d}{dx}\sum_{n=0}^{\infty} x^n \\ &= \sum_{n=0}^{\infty} nx^{n-1} \end{aligned}$$

so $x/(1-x)^2 = \sum_{n=0}^{\infty} nx^n$. Hence $I = e^{-1}/(1-e^{-1})^2 = e/(e-1)^2$.

Problem 4

Evaluate

$$\int_0^{2026} \left\lceil \frac{\lceil x \rceil}{2} \right\rceil dx$$

Solution 4

We note that if $2k < x < 2k+2$ then the function value at x is $k+1$, so

$$\begin{aligned} I &= \int_0^2 dx + \int_2^4 2dx + \cdots \int_{2024}^{2026} 1013dx \\ &= 2(1 + 2 + \cdots 1013) \\ &= 1013 \cdot 1014 \\ &= 1027182. \end{aligned}$$

Leibniz Rule (Feynman)

Problem 5

Evaluate

$$\int_0^\infty \frac{e^{-x}\sin(67x)}{x}\, dx.$$

Solution 5

Let

$$I(\alpha) = \int_0^\infty \frac{e^{-ax}\sin(\alpha x)}{x} dx$$

such that

$$\begin{aligned} I'(\alpha) &= \int_0^\infty e^{-ax}\cos(\alpha x) dx \\ &= \mathcal{L}\{\cos(\alpha x)\}(a) \\ &= \frac{a}{a^2 + \alpha^2} \end{aligned}$$

and set $a = 1$ so $I(\alpha) = \tan^{-1}(\alpha)$, as $I(0) = 0$. Hence $I = I(67) = \tan^{-1}(67)$.

Problem 6

Evaluate

$$\int_0^\infty \frac{\ln(1+4x^2)}{1+x^2}\, dx.$$

Solution 6

Let

$$I(a) = \int_0^\infty \frac{\ln(1+a^2x^2)}{1+x^2}dx$$

such that

$$\begin{aligned} I'(\alpha) &= \int_0^\infty \frac{2ax^2}{(1+x^2)(1+a^2x^2)}dx \\ &= \frac{2a}{a^2-1}\int_0^\infty \frac{1}{x^2+1} - \frac{1}{1+a^2x^2}dx \\ &= \frac{2a}{a^2-1}\left(\frac{\pi}{2} - \frac{\pi}{2a}\right) \\ &= \frac{\pi}{a+1} \end{aligned}$$

so $I(\alpha) = \pi \ln|a+1|$, as $I(0) = 0$. Hence $I = I(2) = \pi \ln(3)$.

Gamma and Beta Functions

Problem 7

Recall that $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}\,dt$. Verify that, for $n \in \mathbb{N}$, $\Gamma(n+1) = n!$. Hence, evaluate $\int_0^1 \left(\ln\left(\frac{1}{x}\right)\right)^{a-1}\,dx$, for $a \in \mathbb{Z}^+$.

Solution 7

We can prove the identity by induction. First, observe that

$$\Gamma(1) = \int_0^\infty e^{-t}\,dt = 1 = 0!.$$

For the inductive step, suppose that $\Gamma(n) = (n-1)!$. Then

$$\begin{aligned} \Gamma(n+1) &= \int_0^\infty t^n e^{-t}\,dt \\ &= \left[-e^{-t}t^n\right]_0^\infty + n\int_0^\infty t^{n-1}e^{-t}\,dt \\ &= n\Gamma(n) \\ &= n!. \end{aligned}$$

To evaluate the integral, let

$$u = \ln\left(\frac{1}{x}\right) = -\ln x.$$

Then $x = e^{-u}$ and $dx = -e^{-u}\,du$. Therefore, for $a > 0$,

$$\begin{aligned} \int_0^1 \left(\ln\left(\frac{1}{x}\right)\right)^{a-1}dx &= \int_0^\infty u^{a-1}e^{-u}\,du \\ &= \Gamma(a) = a!. \end{aligned}$$

Problem 8

Recall that

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx, \qquad m,n>0, \qquad B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$$

For $a,b \in \mathbb{R}$ with $a<b$ and $m,n \in \mathbb{Z}^+$, find

$$\int_a^b (b-x)^{m-1}(x-a)^{n-1}\,dx.$$

Solution 8

Using the substitution $u = \frac{x-a}{b-a}$, we obtain

$$\begin{aligned}
\int_a^b (b-x)^{m-1}(x-a)^{n-1}\,dx &= \int_0^1 (b-a)(b-a)^{m-1}(b-a)^{n-1}(1-u)^{m-1}u^{n-1}\,du \\
&= (b-a)^{m+n-1}\int_0^1 (1-u)^{m-1}u^{n-1}\,du \\
&= (b-a)^{m+n-1}B(n,m) \\
&= (b-a)^{m+n-1}\frac{\Gamma(n)\Gamma(m)}{\Gamma(n+m)} \\
&= (b-a)^{m+n-1}\frac{(n-1)!(m-1)!}{(n+m-1)!}.
\end{aligned}$$

Challenges

Problem 9

Evaluate

$$\int_0^\infty \frac{1}{(1+x^\varphi)^\varphi}dx,$$

where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

Solution 9

We make the substitution $u = x^\varphi$ such that

$$I = \int_0^\infty \frac{u^{\frac{1}{\varphi}-1}}{\varphi(1+u)^\varphi}du.$$

Recall that

$$B(m,n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx,$$

then we have that

$$\begin{aligned} I &= \frac{1}{\varphi} B\left(\frac{1}{\varphi}, \varphi - \frac{1}{\varphi}\right) \\ &= \frac{1}{\varphi} \frac{\Gamma\left(\frac{1}{\varphi}\right)\Gamma\left(\varphi - \frac{1}{\varphi}\right)}{\Gamma(\varphi)}. \end{aligned}$$

This looks scary, but we know that $\varphi^2 - \varphi - 1 = 0$ because φ is the golden ratio, so actually $\varphi - \frac{1}{\varphi} = 1$ and $\frac{1}{\varphi} = \varphi - 1$ and we get that

$$\begin{aligned} I &= \frac{\Gamma(\varphi - 1)\Gamma(1)}{\varphi\Gamma(\varphi)} \\ &= \frac{\Gamma(\varphi - 1)}{\varphi(\varphi - 1)\Gamma(\varphi - 1)} \\ &= \frac{1}{\varphi^2 - \varphi} \\ &= 1 \end{aligned}$$

using that $x\Gamma(x) = \Gamma(x + 1)$.

Problem 10

Evaluate

$$\int_0^\infty \frac{\tan^{-1}(\pi x) - \tan^{-1}(3x)}{x} dx.$$

Solution 10

Let

$$I(\alpha, \beta) = \int_0^\infty \frac{\tan^{-1}(\alpha x) - \tan^{-1}(\beta x)}{x} dx$$

such that

$$\begin{aligned} I_\alpha(\alpha, \beta) &= \int_0^\infty \frac{1}{1 + \alpha^2 x^2} dx \\ &= \left[\frac{1}{\alpha} \tan^{-1}(\alpha x)\right]_0^\infty \\ &= \frac{\pi}{2\alpha} \end{aligned}$$

and so

$$I(\alpha, \beta) = \frac{\pi}{2} \ln|\alpha| + C(\beta).$$

Since $I(\beta, \beta) = 0$, we have that $C(\beta) = -\pi/2 \ln|\beta|$ and so

$$I(\alpha, \beta) = \frac{\pi}{2}(\ln|\alpha| - \ln|\beta|)$$

hence $I = I(\pi, 3) = \frac{\pi}{2} \ln(\pi/3)$. You may also use Frullani's integral formula, but that wasn't mentioned in the workshop :(

Problem 11

Evaluate

$$\int_0^\infty e^{-x^2}\cos(x)\,dx.$$

Solution 11

Let

$$I(\alpha) = \int_0^\infty e^{-x^2}\cos(\alpha x)dx$$

such that

$$\begin{aligned} I'(\alpha) &= \int_0^\infty -xe^{-x^2}\sin(\alpha x)dx \\ &= \left[\frac{1}{2}e^{-x^2}\sin(\alpha x)\right]_0^\infty - \frac{\alpha}{2}\int_0^\infty e^{-x^2}\cos(\alpha x)dx \\ &= \frac{-\alpha}{2}I(\alpha) \end{aligned}$$

which is a differential equation. Solving this differential equation, we get that $I(\alpha) = Ce^{-\alpha^2/4}$. Now we note that

$$\begin{aligned} I(0) &= \int_0^\infty e^{-x^2}dx \\ &= \frac{\sqrt{\pi}}{2} = C \end{aligned}$$

which is a well known result and so $I(\alpha) = \frac{\sqrt{\pi}}{2}e^{-\alpha^2/4}$. Hence $I = I(1) = \frac{\sqrt{\pi}}{2}e^{-1/4}$.

Problem 12

Find

$$\int_0^1 \frac{1}{x^x}\,dx$$

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Solution 12

Observe that $\frac{1}{x^x} = e^{-x\ln x}$. Using the series expansion for the exponential,

$$e^{-x\ln x} = \sum_{k=0}^\infty \frac{(-x\ln x)^k}{k!}.$$

Therefore,

$$\int_0^1 \frac{1}{x^x}\,dx = \sum_{k=0}^\infty \int_0^1 \frac{(-x\ln x)^k}{k!}\,dx.$$

Now let $u = -\ln x$. Then

$$\begin{aligned}
\int_0^1 \frac{1}{x^x}\,dx &= \sum_{k=0}^{\infty} \int_0^{\infty} e^{-u}\frac{(e^{-u}u)^k}{k!}\,du \\
&= \sum_{k=0}^{\infty} \int_0^{\infty} \frac{e^{-(k+1)u}u^k}{k!}\,du \\
&= \sum_{k=0}^{\infty} \frac{1}{k!(k+1)^{k+1}} \int_0^{\infty} e^{-t}t^k\,dt \qquad (t = (k+1)u) \\
&= \sum_{k=0}^{\infty} \frac{\Gamma(k+1)}{k!(k+1)^{k+1}} \\
&= \sum_{k=0}^{\infty} \frac{1}{(k+1)^{k+1}}.
\end{aligned}$$